RANK-OPTIMAL DIRECTIONAL TAYLOR JETS FOR HIGH-ORDER MIXED DERIVATIVES OF NEURAL SURROGATES

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Abstract. Evaluating one prescribed high-order mixed partial derivative of a neural surrogate is a recurring cost in scientific computing, and the price of an evaluation is set by how many passes through the model it takes. Repeated automatic differentiation is exact but nests, numerical differentiation trades exactness for a step size, and randomized estimators trade it for variance. Taylor-mode automatic differentiation removes the nesting by propagating truncated Taylor coefficients along one direction at a time, which leaves open how many directions a prescribed derivative needs. We settle that question. For a fixed order the directional Taylor coefficients form a homogeneous polynomial in the direction, so extracting one mixed derivative extracts one of its coefficients, and a directional formula is a decomposition of a monomial into powers of linear forms. The shortest formula therefore has as many terms as the Waring rank of that monomial, which is known in closed form over the complex numbers and is attained by a roots-of-unity construction that requires no linear solve. The classical polarization identity is the lowest-order case of the same construction, and every repeated index widens the gap between them. Realized as one batched Taylor jet with a closed-form adjoint, the schedule becomes a derivative backend that is algebraically exact, differentiable in the model parameters, and able to take complex directions. Double-precision tests reproduce nested automatic differentiation for values and for parameter gradients. As a demonstration, physics-informed networks built on the backend train under a common wall-clock budget on high-order and complex-valued benchmarks, and are the most accurate method in every setting tested.

Key words. high-order derivatives, Taylor-mode automatic differentiation, Waring decomposition, apolarity, complex-valued neural networks, physics-informed neural networks

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1. Introduction.

Derivatives of a neural surrogate with respect to its inputs are needed wherever a learned function has to satisfy, or be probed against, a differential relation. Neural representations of the solution of a partial differential equation are trained by penalizing the residual of that equation, so every optimization step differentiates the network at many points [6, 12, 16]. Variational Monte Carlo evaluates a differential operator applied to a neural wavefunction at every sampled configuration [14]. In each case the differentiation sits inside the innermost loop, is repeated for the whole of training, and is applied to the same model at many points at once, so its cost is a first-order concern rather than an implementation detail. The concern sharpens with the order of the derivative: a fourth- or sixth-order operator is far more expensive to evaluate than a gradient, and the gap widens with the depth of the model.

The exact ways of computing such a derivative are long established, and each pays for exactness differently. Symbolic differentiation returns a closed-form expression whose size grows quickly with the order. Numerical differentiation evaluates the model on a stencil and is indifferent to its internal structure, but it introduces a step size, and truncation and cancellation errors that degrade as the order rises; coupling it with exact differentiation recovers part of the accuracy while keeping the locality [3]. Automatic differentiation avoids both drawbacks by differentiating the program itself [5]. Its reverse mode is the standard route to a gradient, and a high-order derivative is usually obtained by applying it repeatedly; each application, however, differentiates the graph built by the previous one, so depth and storage grow

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with the order. Taylor mode avoids that nesting: it propagates truncated Taylor coefficients through every primitive, and one forward pass along a chosen direction returns the directional derivatives of all orders up to the truncation [1, 5]. The cost of Taylor mode is therefore counted in passes, one per direction, which makes the number of directions the quantity worth minimizing.

Where exactness can be surrendered, estimation is cheaper still, and a substantial literature now trades it for variance. Hutchinson-type estimators replace the trace of a derivative tensor by an expectation of quadratic forms, so that a Laplacian costs Hessian-vector products rather than a Hessian [8]. Dimension-sampling estimators evaluate a random subset of the coordinate terms of the operator at each step [10]. Randomized smoothing convolves the model with a Gaussian and uses Stein identities to express derivatives of the smoothed function as expectations of function values, removing the differentiation entirely at the price of a smoothing bandwidth and of a variance that grows as the bandwidth shrinks [7, 9, 19]. The stochastic Taylor derivative estimator generalizes the idea to an arbitrary contraction of a derivative tensor by pushing forward randomly drawn sparse jets [17]. These methods are unbiased and scale to dimensions that exact evaluation cannot reach, but the estimate they return is exact only in expectation, and the sampling budget reappears in the accuracy of whatever is trained on it.

A third line keeps exactness and buys speed from the structure of one particular operator. The forward Laplacian propagates the value, the gradient, and the Laplacian together through the network, exploiting the way a trace collapses through a linear layer, and is now standard in variational Monte Carlo [4, 14]. Specializations of this kind are exact and fast, but each is tied to the operator it was designed for.

What is missing from all three lines is an answer for the elementary exact case: one prescribed mixed partial derivative, computed deterministically, of a model that can be evaluated along any direction. Taylor mode supplies the derivative along whichever direction it is given, and it is silent about how many directions the target requires. The classical answer is the polarization identity, which reconstructs a mixed derivative from signed evaluations along sums of coordinate directions; its count is exponential in the order and depends on the order alone, so it treats a derivative that repeats a few coordinates exactly like one that touches many. That insensitivity is suspicious, since a repeated index carries less information than a square-free one, and it raises a question that seems not to have been asked: for a given mixed derivative, what is the smallest number of directional evaluations that can reconstruct it exactly, and which directions attain it?

This paper answers that question and turns the answer into an implementation. The route is to stop thinking of the directions as a stencil and to look at what the directional Taylor coefficients are as a function of the direction: for a fixed order they form a homogeneous polynomial, and extracting one mixed derivative is extracting one of its coefficients. A directional formula is then a decomposition of a monomial into powers of linear forms, so the shortest formula has as many terms as the Waring rank of that monomial, a quantity the algebraic-geometry literature settled in closed form. Over the complex numbers the minimizing directions are roots of unity and are written down without solving anything, the classical polarization identity reappears as the lowest-order instance of the same construction, and every repeated index widens the gap between them. Because the construction is exact rather than approximate, it composes with the training loop in the ordinary way: we evaluate the whole schedule in one batched Taylor jet through an entire activation, and give the jet a closed-form adjoint so that the result stays differentiable in the model parameters.

The contributions of this work are as follows:

(a) We prove that directional formulas for a fixed mixed derivative correspond exactly to Waring decompositions of the associated monomial, so that the minimum number of directional evaluations is a Waring rank, and we give a schedule that attains it in closed form.

(b) We realize that schedule as a single batched Taylor jet with a closed-form adjoint, obtaining a derivative backend that is algebraically exact, differentiable in the model parameters, admits complex directions natively, and agrees with nested automatic differentiation in double precision.

(c) We demonstrate the backend inside complete physics-informed training runs on high-order and complex-valued benchmarks under a common wall-clock budget, reporting accuracy, throughput, and trainable degrees of freedom.

The remainder of this paper is organized as follows. Section 2 fixes the evaluation problem, reviews Taylor-mode evaluation of directional derivatives, and recalls the polarization identity that our construction replaces. Section 3 identifies the shortest directional formula with a monomial Waring decomposition, states and proves the main theorem, and gives the resulting algorithm. Section 4 verifies the implementation against nested automatic differentiation and then demonstrates it inside complete training runs on three benchmark families. Section 5 concludes. The roots-of-unity filter, the explicit schedules used in the experiments, and the apolar form of the rank bound are collected in Appendix A.

2. Preliminaries.

2.1. The evaluation problem. Let $u_\theta \colon \mathbb{R}^d \to \mathbb{R}$ be a neural surrogate with parameters θ, smooth in its inputs, and let ν be a multi-index of order p. The task addressed in this paper is the evaluation of the single mixed partial derivative $\partial^\nu u_\theta$ at a given point, subject to three requirements that together rule out most of the alternatives surveyed in Section 1. The value must be exact, so that no step size and no sampling budget enters the result. The multi-index is fixed and prescribed, rather than being contracted against a structured family such as a trace, so an operator-specific specialization does not apply. And the value must remain differentiable in θ, since it is consumed by an optimizer that updates the surrogate. The derivative is also evaluated at many points at once and re-evaluated at every step, so what matters is the cost of one evaluation of one derivative, amortized over a batch.

The benchmarks of Section 4 instantiate this task in its most demanding common form, the residual of a boundary-value problem. Let $\Omega \subset \mathbb{R}^d$ be a bounded domain with boundary $\Gamma := \partial\Omega$, and consider

$$\mathcal{P}u = f \quad \text{in } \Omega, \qquad \mathcal{B}u = g \quad \text{on } \Gamma, \tag{2.1}$$

with $\mathcal{P}$ a differential operator of order p and $\mathcal{B}$ the boundary conditions. Given interior points $(\boldsymbol{x}_{r,i})_{i=1}^{n_r} \subset \Omega$ and boundary points $(\boldsymbol{x}_{b,i})_{i=1}^{n_b} \subset \Gamma$, a physics-informed network fits (2.1) by minimizing

$$\mathcal{L}(\theta) := \frac{\lambda_r}{n_r} \sum_{i=1}^{n_r} \left|\mathcal{P}u_\theta(\boldsymbol{x}_{r,i}) - f(\boldsymbol{x}_{r,i})\right|^2 + \frac{\lambda_b}{n_b} \sum_{i=1}^{n_b} \left|\mathcal{B}u_\theta(\boldsymbol{x}_{b,i}) - g(\boldsymbol{x}_{b,i})\right|^2 . \tag{2.2}$$

Expanding $\mathcal{P}$ into monomial terms turns one evaluation of the residual in (2.2) into several instances of the task above, on the same surrogate and the same batch of points; a polyharmonic operator $\mathcal{P} = \Delta^m$, for example, expands into $\binom{d+m-1}{m}$ of them. The loss (2.2) is a consumer of the derivative and not the object of study: nothing in what follows is specific to it.

2.2. Taylor-mode evaluation of directional derivatives. Let $u\colon \mathbb{R}^d \to \mathbb{R}$ be sufficiently smooth near a fixed point $\boldsymbol{x}$. For a direction $\boldsymbol{v} \in \mathbb{R}^d$, the directional Taylor coefficient of order p at $\boldsymbol{x}$ along $\boldsymbol{v}$ is

$$T_p(\boldsymbol{x};\boldsymbol{v}) := \frac{1}{p!}\,\frac{d^p}{d\tau^p}\,u(\boldsymbol{x}+\tau\boldsymbol{v})\Big|_{\tau=0} = \frac{1}{p!}\,D^p u(\boldsymbol{x})[\boldsymbol{v},\dots,\boldsymbol{v}]. \tag{2.3}$$

Taylor-mode automatic differentiation evaluates (2.3) in one forward pass [1,5]. Each intermediate activation $y \in \mathbb{K}^B$ over a batch of B points, with $\mathbb{K} \in \{\mathbb{R},\mathbb{C}\}$, is replaced by the length-$(p+1)$ tuple $(y_0,\dots,y_p)$ of its truncated Taylor coefficients $y_k = (1/k!)\,d^k y/d\tau^k|_{\tau=0}$. For a multilayer perceptron $u = \phi_L \circ A_L \circ \cdots \circ \phi_1 \circ A_1$ with affine maps A_ℓ and entrywise activations ϕ_ℓ, an affine layer $y = Wx + b$ propagates each order independently,

$$y_0 = Wx_0 + b, \qquad y_k = Wx_k \quad (k \ge 1), \tag{2.4}$$

and each nonlinearity is propagated together with one auxiliary jet that closes its derivative recurrence. For tanh the auxiliary jet is $z = 1 - y^2$, initialized by $y_0 = \tanh(x_0)$ and $z_0 = 1 - y_0^2$, and

$$y_k = \frac{1}{k}\sum_{j=0}^{k-1}(k-j)\,z_j\,x_{k-j}, \qquad z_k = -\sum_{j=0}^{k} y_j\,y_{k-j}, \qquad k \ge 1. \tag{2.5}$$

For sinh the auxiliary jet is $z = \cosh(x)$, initialized by $y_0 = \sinh(x_0)$ and $z_0 = \cosh(x_0)$, and

$$y_k = \frac{1}{k}\sum_{j=0}^{k-1}(k-j)\,z_j\,x_{k-j}, \qquad z_k = \frac{1}{k}\sum_{j=0}^{k-1}(k-j)\,y_j\,x_{k-j}, \qquad k \ge 1. \tag{2.6}$$

The recurrence (2.6) has the same coupled form as the one for sin and cos, except that the cosh update carries no sign flip. The implementation also contains the recurrences for sin and for the logistic sigmoid, which the benchmark architectures of Section 4 require. Every coefficient y_k is an ordinary tensor, and the propagation stores $p+1$ coefficient tensors per activation without nesting input-gradient graphs.

2.3. Polarization: a second-order warm-up. Throughout the paper, $\nu = (\nu_1,\dots,\nu_d) \in \mathbb{Z}^d_{\ge 0}$ denotes a multi-index of order $p = |\nu|$, and $\partial^\nu := \partial_1^{\nu_1}\cdots\partial_d^{\nu_d}$ denotes the associated derivative. The classical route from directional coefficients to ∂^ν is the polarization identity. Writing the differentiated coordinates as a list $i_1,\dots,i_p$ in which coordinate k occurs ν_k times, it reads

$$\partial^\nu u(\boldsymbol{x}) = \frac{1}{2^p}\sum_{\epsilon\in\{\pm1\}^p}\Big(\textstyle\prod_{r=1}^p \epsilon_r\Big)\,T_p\Big(\boldsymbol{x};\,\textstyle\sum_{r=1}^p \epsilon_r e_{i_r}\Big). \tag{2.7}$$

The smallest instance is instructive. For $p = 2$ and $\partial^\nu = \partial_1\partial_2$, expanding $T_p(\boldsymbol{x};\boldsymbol{v}) = \frac{1}{2}(u_{11}v_1^2 + 2u_{12}v_1v_2 + u_{22}v_2^2)$ gives $T_p(\boldsymbol{x}; e_1 \pm e_2) = \frac{1}{2}(u_{11} \pm 2u_{12} + u_{22})$, so the two probes $e_1 + e_2$ and $e_1 - e_2$ suffice:

$$\partial_1\partial_2 u(\boldsymbol{x}) = \tfrac{1}{2}T_p(\boldsymbol{x}; e_1 + e_2) - \tfrac{1}{2}T_p(\boldsymbol{x}; e_1 - e_2). \tag{2.8}$$

The diagonal terms cancel in the difference, and this cancellation is the entire mechanism of (2.7).

Beyond second order the identity is no longer economical. The sum in (2.7) has 2^p terms; duplicate directions arising from repeated coordinates and antipodal pairs can be merged, which leaves at most 2^{p-1} distinct probes, and a further merge of proportional directions is possible because $T_p(\boldsymbol{x};\lambda\boldsymbol{v}) = \lambda^p T_p(\boldsymbol{x};\boldsymbol{v})$. The resulting count is still exponential in p, and, more importantly, it barely responds to the structure of ν: the sixth-order derivative $\partial_1^2\partial_2^2\partial_3^2$ and the square-free $\partial_1\partial_2\cdots\partial_6$ receive 13 and 32 probes respectively, although the first involves only three coordinates. Section 3 shows that 9 probes are necessary and sufficient in the first case, and that 32 cannot be improved in the second.

3. Rank-optimal directional schedules. We introduce some notation used throughout this section. Let $\mathbb{N}_0 := \{0, 1, 2, \ldots\}$, and let $\mathbb{C}^d$ be the d-dimensional complex space. The *active coordinates* of ν are the indices k with $\nu_k > 0$; we relabel them as $i_0, \ldots, i_n$ and set $a_j := \nu_{i_j}$, ordered so that $1 \le a_0 \le \cdots \le a_n$, with ties broken by the original coordinate index. The ordering singles out a coordinate i_0 of least exponent, which is the coordinate the rank formula treats separately. We write $a := (a_0, \ldots, a_n)$, and we record for later use that

$$p = \sum_{j=0}^{n} a_j, \qquad \nu! = \prod_{k=1}^{d} \nu_k! = \prod_{j=0}^{n} a_j!, \qquad m_j := a_j + 1. \tag{3.1}$$

The *active subspace* is $V := \operatorname{span}\{e_{i_0}, \ldots, e_{i_n}\} \subset \mathbb{R}^d$, and $V_{\mathbb{C}} := V \otimes_{\mathbb{R}} \mathbb{C}$ is its complexification. Since ∂^ν is insensitive to coordinates outside the active set, all directional probes below may be taken in V or in $V_{\mathbb{C}}$. For $\mathbb{K} \in \{\mathbb{R}, \mathbb{C}\}$, the space of homogeneous polynomials of degree p in $\boldsymbol{z} = (z_0, \ldots, z_n)$ is denoted $\mathbb{K}[\boldsymbol{z}]_p$, and $[\boldsymbol{z}^a]F$ denotes the coefficient of $\boldsymbol{z}^a = z_0^{a_0} \cdots z_n^{a_n}$ in F. For $m \ge 1$, let $U_m := \{\zeta \in \mathbb{C} : \zeta^m = 1\}$, and let $\mathbf{1}_A$ be the indicator of a condition A.

The construction of this section uses complex directions, so we fix their meaning once. For $\boldsymbol{v} \in V_{\mathbb{C}}$ we define $T_p(\boldsymbol{x};\boldsymbol{v})$ by the complex multilinear extension of $D^p u(\boldsymbol{x})$ on the right-hand side of (2.3). When u admits a holomorphic extension to a neighbourhood of $\boldsymbol{x}$ in $\mathbb{C}^d$, this definition agrees with the order-p coefficient of $\tau \mapsto u(\boldsymbol{x}+\tau\boldsymbol{v})$, and the recurrences (2.4)–(2.6) evaluate it on complex tensors. In practice the extension is guaranteed by entire activations such as $\sinh(z) = (e^z - e^{-z})/2$; see Remark 3.5. The distinction matters because an arbitrary C^p function on $\mathbb{R}^d$ need not be defined at complex arguments.

3.1. Basic expansion. Let $\boldsymbol{z} \in \mathbb{C}^{n+1}$ parametrize the active subspace through $\boldsymbol{v}(\boldsymbol{z}) := \sum_{j=0}^{n} z_j e_{i_j}$, and define the *generating polynomial* of u at $\boldsymbol{x}$ by

$$Q_u(\boldsymbol{z}) := T_p\bigl(\boldsymbol{x};\boldsymbol{v}(\boldsymbol{z})\bigr). \tag{3.2}$$

The symmetric p-linearity of $D^p u(\boldsymbol{x})$ and the multinomial theorem give

$$D^p u(\boldsymbol{x})\bigl[\boldsymbol{v}(\boldsymbol{z}), \ldots, \boldsymbol{v}(\boldsymbol{z})\bigr] = \sum_{|\beta|=p} \binom{p}{\beta} \boldsymbol{z}^\beta\, \partial^\beta u(\boldsymbol{x}) = \sum_{|\beta|=p} \frac{p!}{\beta!}\, \boldsymbol{z}^\beta\, \partial^\beta u(\boldsymbol{x}),$$

where β ranges over multi-indices on the active variables and $\partial^\beta := \partial_{i_0}^{\beta_0} \cdots \partial_{i_n}^{\beta_n}$. Dividing by $p!$ and comparing with (2.3) and (3.2) yields

$$Q_u(\boldsymbol{z}) = \sum_{|\beta|=p} \frac{\partial^\beta u(\boldsymbol{x})}{\beta!}\, \boldsymbol{z}^\beta, \tag{3.3}$$

so Q_u is homogeneous of degree p and its coefficients are the order-p derivatives of u at $\boldsymbol{x}$. Reading off the coefficient of $\boldsymbol{z}^a$ in (3.3) and using $a! = \nu!$ from (3.1) gives

$$\partial^\nu u(\boldsymbol{x}) = \nu!\,[\boldsymbol{z}^a]\,Q_u(\boldsymbol{z}). \tag{3.4}$$

Extracting one mixed derivative is therefore the extraction of one coefficient of a homogeneous polynomial of degree p.

The evaluations available to us are values of Q_u at individual directions, so we ask for the shortest linear combination of such values that realizes the functional (3.4). A *directional schedule* of length R for ∂^ν over $\mathbb{K}$ is a finite collection $(c_r, \boldsymbol{v}_r)_{r=1}^R \subset \mathbb{K}\times V_{\mathbb{K}}$ such that

$$\partial^\nu u(\boldsymbol{x}) = \sum_{r=1}^{R} c_r\, T_p(\boldsymbol{x}; \boldsymbol{v}_r) \tag{3.5}$$

for every u that is C^p on a neighbourhood of $\boldsymbol{x}$, and, when $\mathbb{K} = \mathbb{C}$, holomorphic there. Because (3.5) is required for all such u, the order-p derivatives $(\partial^\beta u(\boldsymbol{x}))_{|\beta|=p}$ may be prescribed independently: for any coefficients q_β, the polynomial $\sum_\beta q_\beta \prod_j (y_{i_j} - x_{i_j})^{\beta_j}$ has $\partial^\beta u(\boldsymbol{x}) = \beta!\, q_\beta$. Substituting (3.3) into the right-hand side of (3.5) gives

$$\sum_{r=1}^{R} c_r\, T_p(\boldsymbol{x}; \boldsymbol{v}_r) = \sum_{|\beta|=p} \frac{\partial^\beta u(\boldsymbol{x})}{\beta!} \Big(\sum_{r=1}^{R} c_r \boldsymbol{v}_r^\beta\Big), \tag{3.6}$$

and comparing (3.6) with the target $\partial^\nu u(\boldsymbol{x}) = \partial^a u(\boldsymbol{x})$ coefficient by coefficient shows that (3.5) holds for every admissible u if and only if

$$\sum_{r=1}^{R} c_r \boldsymbol{v}_r^\beta = \nu!\,\delta_{a\beta}, \qquad |\beta| = p, \tag{3.7}$$

where $\delta_{a\beta} := 1$ if $\beta = a$ and $\delta_{a\beta} := 0$ otherwise. The design problem is thus to solve the moment conditions (3.7) with as few directions as possible.

REMARK 3.1 (Polarization is far from minimal). *The merged polarization identity of Section 2.3 solves* (3.7) *with at most* 2^{p-1} *real directions, a count that depends on* p *alone. The moment conditions, by contrast, are only* $\binom{p+n}{n}$ *equations on the* $n+1$ *active variables, and their number is governed by the number of active coordinates rather than by* p. *For a derivative such as* $\partial_1^2\partial_2^2\partial_3^2$, *which has* $p = 6$ *but only three active coordinates, the gap between the two counts is large, and Theorem 3.3 shows how to close it.*

3.2. Minimal schedules and monomial Waring rank. Pair a direction $\boldsymbol{v} \in V_{\mathbb{C}}$, written $\boldsymbol{v} = \sum_{j=0}^n v_j e_{i_j}$ in active coordinates, with the polynomial variable $\boldsymbol{z}$ through $\boldsymbol{v}\cdot\boldsymbol{z} := \sum_{j=0}^n v_j z_j$. Expanding the pth power of this pairing by the multinomial theorem gives

$$\sum_{r=1}^{R} c_r (\boldsymbol{v}_r\cdot\boldsymbol{z})^p = \sum_{|\beta|=p} \binom{p}{\beta} \Big(\sum_{r=1}^{R} c_r \boldsymbol{v}_r^\beta\Big) \boldsymbol{z}^\beta, \tag{3.8}$$

whose coefficients are the very quantities constrained by (3.7). The moment conditions are therefore a polynomial identity in disguise, and the shortest schedule is the shortest decomposition of a monomial into pure powers of linear forms. That notion has a classical name.

DEFINITION 3.2 (Waring rank). *Let $\mathbb{K} \in \{\mathbb{R}, \mathbb{C}\}$ and let $F \in \mathbb{K}[\boldsymbol{z}]_p$ be homogeneous of degree p. The* Waring rank $R_{\mathbb{K}}(F)$ *is the least R for which there exist $c_r \in \mathbb{K}$ and linear forms $L_r \in \mathbb{K}[\boldsymbol{z}]_1$ with $F = \sum_{r=1}^{R} c_r L_r(\boldsymbol{z})^p$.*

Since $\mathbb{R} \subset \mathbb{C}$, every real decomposition is a complex one, so $R_{\mathbb{C}}(F) \leq R_{\mathbb{R}}(F)$, and enlarging the field can only shorten a schedule. The following theorem is the main result of the paper. Its three parts identify the design problem with a Waring decomposition, evaluate the resulting minimum, and exhibit a schedule of that length in closed form.

THEOREM 3.3 (Rank-optimal directional schedule). *Let ν be a multi-index of order $p \geq 1$ with active exponents $1 \leq a_0 \leq \cdots \leq a_n$ and $m_j := a_j + 1$ as in* (3.1), *and let $\mathbb{K} \in \{\mathbb{R}, \mathbb{C}\}$.*

(i) *A collection $(c_r, \boldsymbol{v}_r)_{r=1}^{R} \subset \mathbb{K} \times V_{\mathbb{K}}$ is a directional schedule for ∂^ν if and only if*

$$p!\,\boldsymbol{z}^a = \sum_{r=1}^{R} c_r (\boldsymbol{v}_r \cdot \boldsymbol{z})^p \qquad \text{in } \mathbb{K}[\boldsymbol{z}]_p. \tag{3.9}$$

(ii) *The minimum length of a directional schedule for ∂^ν over $\mathbb{K}$ equals $R_{\mathbb{K}}(\boldsymbol{z}^a)$, and over $\mathbb{C}$ this minimum is*

$$R_{\mathbb{C}}(\boldsymbol{z}^a) = \prod_{j=1}^{n} m_j = \frac{\prod_{j=0}^{n} m_j}{m_0}. \tag{3.10}$$

(iii) *For $\zeta = (\zeta_1, \ldots, \zeta_n) \in U_{m_1} \times \cdots \times U_{m_n}$, put*

$$\boldsymbol{v}_\zeta := e_{i_0} + \sum_{j=1}^{n} \zeta_j e_{i_j}, \qquad c_\zeta := \frac{\nu!}{\prod_{j=1}^{n} m_j} \prod_{j=1}^{n} \zeta_j. \tag{3.11}$$

Then $(c_\zeta, \boldsymbol{v}_\zeta)_\zeta$ is a directional schedule for ∂^ν of length $\prod_{j=1}^{n} m_j$, and by part (ii) no complex schedule is shorter.

Proof.

(i) The moment conditions are a polynomial identity. By (3.7), the collection is a directional schedule if and only if $\sum_r c_r \boldsymbol{v}_r^\beta = \nu!\,\delta_{a\beta}$ for every β with $|\beta| = p$. By (3.8), the identity (3.9) holds if and only if $\binom{p}{\beta} \sum_r c_r \boldsymbol{v}_r^\beta = p!\,\delta_{a\beta}$ for every such β. Dividing the second family of conditions by $\binom{p}{\beta}$ turns it into $\sum_r c_r \boldsymbol{v}_r^\beta = \beta!\,\delta_{a\beta}$, and $\beta!\,\delta_{a\beta} = \nu!\,\delta_{a\beta}$ by (3.1). The two families coincide, which proves part (i).

(ii) The minimum length is a Waring rank. Absorbing the nonzero constant $p!$ into the coefficients c_r shows that a decomposition of $p!\,\boldsymbol{z}^a$ into R pure pth powers exists if and only if one of $\boldsymbol{z}^a$ does, so $R_{\mathbb{K}}(p!\,\boldsymbol{z}^a) = R_{\mathbb{K}}(\boldsymbol{z}^a)$. Combining this with part (i) identifies the minimum schedule length over $\mathbb{K}$ with $R_{\mathbb{K}}(\boldsymbol{z}^a)$. The value (3.10) of the complex rank of a monomial is the theorem of Carlini, Catalisano, and Geramita [2]; Appendix A.3 recalls the apolarity argument behind its lower bound.

(iii) The roots-of-unity schedule attains the minimum. We use the conventions that an empty product equals 1 and that a Cartesian product over an empty index set has one element, the empty tuple; these cover the case $n = 0$, in which (3.11) produces the single direction e_{i_0} with coefficient $\nu!$. By part (i) it suffices to prove the identity $\sum_\zeta c_\zeta(\boldsymbol{v}_\zeta \cdot \boldsymbol{z})^p = p!\,\boldsymbol{z}^a$. Fix β with $|\beta| = p$ and let C_β denote the coefficient of $\boldsymbol{z}^\beta$ on the left-hand side. Inserting (3.11) and applying the multinomial theorem gives

$$C_\beta = \frac{\nu!}{\prod_{j=1}^n m_j}\binom{p}{\beta}\sum_{\zeta_1\in U_{m_1}}\cdots\sum_{\zeta_n\in U_{m_n}}\prod_{j=1}^n \zeta_j^{\beta_j+1} = \frac{\nu!}{\prod_{j=1}^n m_j}\binom{p}{\beta}\prod_{j=1}^n\Big(\sum_{\zeta_j\in U_{m_j}}\zeta_j^{\beta_j+1}\Big), \tag{3.12}$$

where the second equality separates the independent sums. By Lemma A.1, the jth factor of the product in (3.12) vanishes unless $m_j \mid \beta_j + 1$, and $m_j = a_j + 1$ turns this divisibility into the congruence $\beta_j \equiv a_j \pmod{m_j}$. Suppose $C_\beta \neq 0$, and write $\beta_j = a_j + t_j m_j$ with $t_j \in \mathbb{N}_0$ for $j = 1, \ldots, n$. The degree constraint $|\beta| = p = \sum_{j=0}^n a_j$ then forces $\beta_0 = a_0 - \sum_{j=1}^n t_j m_j$. If some t_j were positive, then $m_j = a_j + 1 \geq a_0 + 1$ by the ordering of the exponents, whence $\beta_0 \leq a_0 - m_j < 0$, contradicting $\beta_0 \geq 0$. Every t_j therefore vanishes, so $\beta_j = a_j$ for $j \geq 1$, and the degree constraint gives $\beta_0 = a_0$ as well. Hence $C_\beta = 0$ for every $\beta \neq a$. For $\beta = a$ each inner sum in (3.12) equals m_j, the product cancels the prefactor, and

$$C_a = \nu!\binom{p}{a} = \Big(\prod_{j=0}^n a_j!\Big)\frac{p!}{\prod_{j=0}^n a_j!} = p!\,.$$

The left-hand side of the identity therefore equals $p!\,\boldsymbol{z}^a$, and the schedule has $|U_{m_1} \times \cdots \times U_{m_n}| = \prod_{j=1}^n m_j$ directions. ∎

Theorem 3.3 converts a question about derivative evaluation into a rank computation whose answer is a product of the incremented exponents, and it delivers the minimizing directions without solving a linear system: the coefficients in (3.11) are roots of unity scaled by $\nu!/R_{\mathbb{C}}(\boldsymbol{z}^a)$, so their magnitudes are known in advance. Three consequences deserve separate mention.

Remark 3.4 (Polarization as a special case). *For $\partial^\nu = \partial_1\partial_2$ we have $n = 1$, $a_0 = a_1 = 1$, $m_1 = 2$, and $U_2 = \{+1, -1\}$, so* (3.11) *returns the two pairs $(\frac{1}{2}, e_1 + e_2)$ and $(-\frac{1}{2}, e_1 - e_2)$. This is precisely the two-point identity* (2.8)*, so the second-order polarization warm-up of Section 2.3 is the case $p = n+1 = 2$ of the construction, and it is already rank-optimal. The construction departs from polarization exactly when some exponent exceeds* 1.

Remark 3.5 (Entire activations are essential). *The directions* (3.11) *are complex whenever $m_j \geq 3$ for some j, so the network is evaluated at complex arguments and its activations must admit a holomorphic extension there. The functions* sinh *and* cosh *are entire, and the recurrence* (2.6) *is an algebraic identity that holds verbatim on complex tensors. By contrast* tanh *is meromorphic, with poles at $\mathrm{i}\pi/2 + \mathrm{i}k\pi$ for $k \in \mathbb{Z}$, so its jet is defined only away from those points and the admissible directions depend on the activation input. Entire activations are therefore a requirement of the construction rather than an incidental convenience, and they are the reason the demonstration of Section 4 uses a* sinh *network for the proposed method.*

Remark 3.6 (Scope of the rank reduction). *Theorem 3.3 concerns one monomial derivative at a time. For the square-free multi-index $a = (1, \ldots, 1)$ of order p the rank* (3.10) *equals 2^{p-1}, which is exactly the merged polarization count, so the construction saves nothing. An operator sum such as Δ^m must be expanded and evaluated term by term, and the theorem then applies to each term separately; it makes no claim of joint optimality for the sum, and a trace-collapsing method [4, 14] can be* *preferable when only a Laplacian is required. The stochastic Taylor derivative estimator [17] solves the strictly broader problem of an arbitrary contraction of the order-p derivative tensor, with unbiased estimates whose variance is controlled by a sampling budget; the two approaches are complementary, and the schedule of Theorem 3.3 is the deterministic minimum for one fixed multi-index.*

Table 3.1 lists the counts produced by the three rules for the active-exponent patterns up to $p = 6$. The complex rank never exceeds a real count, and it falls strictly below the projectively consolidated count exactly for the three patterns in which at least two coordinates carry a repeated exponent. The explicit schedules for the patterns used in Section 4 are listed in Appendix A.2.

TABLE 3.1
Directional counts for representative active-exponent patterns. $R_{\mathbb{C}}$ is the rank (3.10)*; R_{B2} is the implemented rule that merges duplicate and antipodal integer directions in* (2.7)*; R_{proj} additionally merges proportional directions. Bold complex ranks are strictly smaller than R_{proj}. The square-free pattern (1^p) receives no reduction.*

pattern	example multi-index	$R_{\mathbb{C}}$	R_{B2}	R_{proj}
(p)	$\partial_1^p u$	1	$\lceil p/2 \rceil$	1
$(2,1)$	u_{112}	3	3	3
$(1,1,1)$	u_{123}	4	4	4
$(3,1)$	u_{1112}	4	4	4
$(2,2)$	u_{1122}	**3**	4	4
$(2,1,1)$	u_{1123}	6	6	6
$(1,1,1,1)$	u_{1234}	8	8	8
$(4,2)$	u_{111122}	**5**	7	6
$(2,2,2)$	u_{112233}	**9**	13	13
(1^6)	u_{123456}	32	32	32

3.3. Algorithm, gradients, and cost. Algorithm 1 assembles the pieces. The schedule (3.11) is built once, its directions are stacked into a single batch, and one jet pass through the network returns the order-p coefficient for every direction at once; the reported value is then the coefficient-weighted sum. The schedule tensors depend on ν, the device, and the scalar type, but not on the trainable parameters. The reference implementation builds them at call entry and includes that work in the measured throughput of Section 4, while a caller that reuses one multi-index may cache them. For a real-valued model the conjugate pairing of the roots of unity makes the returned value real in exact arithmetic; the implementation keeps the complex tensor and takes its real part only when the target PDE is real-valued. Replacing lines 2–8 by the merged polarization schedule of Section 2.3 yields the real-direction backend against which we compare.

Algorithm 1 must be differentiable in θ, since its output enters the residual term of (2.2). Differentiating the activation recurrences (2.5)–(2.6) by reverse-mode automatic differentiation would rebuild inside the backward pass the nesting that the jet was introduced to avoid. We therefore register each activation jet as a single primitive with the closed-form adjoint

$$g_x^{(m)} = \sum_{j=0}^{p-m} \overline{q_j}\, g_y^{(m+j)}, \qquad q(\tau) := \phi'\big(x(\tau)\big), \tag{3.13}$$

in which q_j is the jth Taylor coefficient of q and $g^{(m)}$ denotes the cotangent of the order-m coefficient. Equation (3.13) follows from the chain rule applied to the jet of

Algorithm 1 Single-monomial $\partial^\nu u(\boldsymbol{x})$ from a rank-optimal schedule and one batched Taylor jet.

Require: scalar network u built from supported analytic primitives; point $\boldsymbol{x} \in \mathbb{R}^d$; multi-index ν of order p
Ensure: the value $\partial^\nu u(\boldsymbol{x})$

Relabel the active coordinates of ν as $i_0, \dots, i_n$ with exponents $1 \le a_0 \le \cdots \le a_n$.

Set $m_j := a_j + 1$ for $j = 1, \dots, n$ and $\mathcal{I} := U_{m_1} \times \cdots \times U_{m_n}$.
Initialize $\mathcal{S} \leftarrow \emptyset$.
for $\zeta = (\zeta_1, \dots, \zeta_n) \in \mathcal{I}$ **do**
  Set $\boldsymbol{v}_\zeta \leftarrow e_{i_0} + \sum_{j=1}^n \zeta_j e_{i_j}$ and $c_\zeta \leftarrow \nu!\big(\prod_{j=1}^n m_j\big)^{-1} \prod_{j=1}^n \zeta_j$.
  Update $\mathcal{S} \leftarrow \mathcal{S} \cup \{(c_\zeta, \boldsymbol{v}_\zeta)\}$.
end for
Stack the directions of $\mathcal{S}$ into $V_{\mathcal{S}} \in \mathbb{C}^{|\mathcal{S}| \times d}$.
Form the batched input jet $\mathbf{J}_0 \leftarrow (\boldsymbol{x} \text{ replicated}, V_{\mathcal{S}}, 0, \dots, 0)$.
Propagate $\mathbf{J}_0$ through u by (2.4)–(2.6) to obtain $\mathbf{J}_L$.
Set $t \leftarrow [\mathbf{J}_L]_p$, one order-$p$ coefficient per direction.
return $\sum_{r=1}^{|\mathcal{S}|} c_r t_r$.

ϕ and uses the convention of a real-valued loss for complex parameters, so that the conjugation is the Hermitian adjoint of multiplication by q_j. The backward pass thus costs one more triangular convolution per activation and never differentiates through the recurrence itself.

The cost of Algorithm 1 separates into a per-direction and a per-schedule factor. For one direction, the jet stores $p+1$ coefficient tensors per activation, an affine layer applies W to each of them, and the activation recurrences require $\sum_{k=1}^p O(k) = O(p^2)$ elementwise products per scalar. The total work and the directional storage then scale linearly in the schedule length, which is $R_{\mathbb{C}}(\boldsymbol{z}^a)$ for Algorithm 1 and the merged count R_{B2} of Table 3.1 for the polarization backend. The rank reduction is an exact statement about that factor and not about wall-clock time, which also depends on complex arithmetic, kernel launch overhead, memory traffic, and whether parameter gradients are required; for small networks or small batches these terms can dominate. Section 4 therefore reports measured throughput alongside accuracy.

Two properties of the construction bear on its numerical behaviour. First, the schedule coefficients are known in closed form, so no ill-conditioned linear system is solved to obtain them, and the method has neither a finite-difference step size nor a Monte Carlo variance. Second, the remaining error is ordinary floating-point round-off from the Taylor recurrences, from complex arithmetic, and from the final summation over directions. Avoiding a linear solve does not by itself eliminate cancellation, factorial growth of the coefficients, or overflow of sinh at large arguments; the verification of Section 4.1 covers double precision through order six and is not a general conditioning study.

4. Numerical experiments. This section reports two complementary studies. The first verifies the backend itself: that the schedule, the derivative values, and the parameter gradients agree with nested automatic differentiation in double precision, with no PDE and no training involved. The second is a demonstration rather than a further test of the derivative: the backend is placed inside complete physics-informed training runs and asked to sustain a competitive workflow on high-order and complex-valued problems under a fixed compute budget. A demonstration of that kind is the honest way to report the benefit, because a reduction in directional evaluations is an algebraic statement whose effect on a real workload also depends on complex arithmetic, batching, and the backward pass. We therefore compare a complex-parameter sinh network against three real high-frequency baselines on polyharmonic, radial-chirp, and time-harmonic Maxwell problems, and report accuracy together with measured throughput.

4.1. Verification of the derivative backend. All checks use double precision and are executable tests in the repository. The roots-of-unity coefficient filter of Lemma A.1 is evaluated against every degree-four monomial in five variables for the active-exponent pattern $(2, 1, 1)$, with a maximum admissible absolute coefficient error of 10^{-12}. The direction counts for all patterns of Table 3.1 are checked against the closed-form rank (3.10) and against the merging implementation.

The two jet backends are then compared with nested automatic differentiation on real-parameter networks using the tanh, logistic, sin, and sinh activations, for the patterns $(2, 2)$, $(1, 1, 1)$, and (3). Values must agree to relative tolerance 2×10^{-10} and absolute tolerance 2×10^{-11}, and the imaginary remainder of a real target must not exceed 2×10^{-11}. Gradients of a squared derivative loss with respect to every network parameter are compared in the same way, with tolerances 2×10^{-9} relative and 2×10^{-11} absolute, for both real- and complex-parameter networks; additional analytic checks cover the orders 1, 2, 3, 4, and 6 for a scalar complex sinh network. Every check passes at the tolerance stated for it. These tests establish that the implementation computes what Theorem 3.3 predicts; they do not turn a reduction in direction count into a hardware-independent timing claim.

4.2. Shared protocol. A *setting* is one problem together with one value of its sweep parameter, and the study below covers 12 settings. The proposed method is a complex-parameter sinh network, denoted Complex sinh, and the three baselines are SIREN [18], a multiscale Fourier-feature PINN [20, 21] denoted mFF-PINN, and MscaleDNN-2-sin [15]. The three baseline implementations were traced to pinned upstream commits and tested for architecture and derivative fidelity before use.

Protocol `jsc_v2` grants a wall-clock budget of 1200 seconds to each triple of setting, method, and seed, and uses five seeds, four trainable hidden layers, 4096 fixed interior points, 512 fixed boundary points, paired per-seed collocation, and a common held-out set of 8192 points. Every hidden layer has width $H = 128$. This shared-width control does not equalize trainable parameter counts: each complex parameter counts as two real degrees of freedom, frozen Fourier maps are excluded, and a split-real Maxwell model contains two width-128 networks. The resulting counts are reported with each table.

All methods use Adam with initial learning rate 10^{-3}, cosine decay to 10^{-4}, and the same residual normalization and boundary penalty within a setting, and the four methods receive identical collocation points for a given seed. Complex sinh uses `complex128` parameters; a real-valued target is fitted through its real output component, with a 10^{-6} penalty on the imaginary parts of the parameters. For Maxwell the full complex output is fitted, and each real baseline represents the real and imaginary components by two separate networks. For the polyharmonic problems the Complex sinh first-layer scale is 2π and the mFF-PINN branches use the scales 1 and π; for the chirp and Maxwell problems these become $\max(10, 2\pi a)$ and $(1, \max(2, \pi a))$, respectively. SIREN retains its upstream first- and hidden-layer frequencies of 30, and

MscaleDNN-2-sin uses three independent subnets at the fixed input scales 1, 2, and 4.

All four architectures use Algorithm 1 for every monomial derivative. Scaled activations and branched embeddings are admitted only after their input derivatives and parameter gradients agree with direct automatic differentiation. Accuracy is the held-out relative L^2 error

$$\mathcal{E}_{L^2} := \frac{\|u_\theta - u_\star\|_{L^2(\Omega)}}{\|u_\star\|_{L^2(\Omega)}}, \tag{4.1}$$

and tables report its arithmetic mean and sample standard deviation over the five seeds, while history panels show the geometric mean and the seedwise range. Each history panel is framed on the seed-mean curves, so a seedwise band leaves the panel wherever an individual seed diverges. A setting is reported only after automated checks confirm all 20 method-seed rows, that is, four methods and five seeds, together with finite metrics, nonempty monotone histories, the width and protocol metadata, and a fixed evaluation rule. All runs used one NVIDIA H20 GPU with CUDA 12.1; the stored records specify PyTorch 2.4.1 or 2.5.1, so throughput is compared within a setting and not across settings.

4.3. Polyharmonic equations. For $d \in \{2,3\}$ we solve the manufactured polyharmonic problem

$$\Delta^m u = (-d\pi^2)^m u_\star \quad \text{in } (-1,1)^d, \qquad u_\star(\boldsymbol{x}) = \prod_{j=1}^{d} \sin(\pi x_j), \tag{4.2}$$

at the orders $2m \in \{2,4,6\}$, with the Navier boundary targets $u = u_\star$ and $\Delta^j u = (-d\pi^2)^j u_\star$ for $j = 1,\dots,m-1$. All of these targets vanish for the chosen $u_\star$. Expanding Δ^m produces $\binom{d+m-1}{m}$ monomial terms, each evaluated independently: a mixed term such as $\partial_1^2\partial_2^2 u$ uses a schedule of length 3 by (3.10), whereas a pure power $\partial_1^{2m} u$ needs one direction.

Table 4.1

Polyharmonic benchmark, $d = 2$: mean $\pm$ sample standard deviation of the held-out relative L^2 error over five seeds after 1200 s. The lowest mean in each row is set in bold.

$2m$	Complex sinh	SIREN	mFF-PINN	MscaleDNN-2-sin
2	$\mathbf{3.36 \pm 1.57} \times 10^{-4}$	1.00 ± 0.00	$4.88 \pm 2.73 \times 10^{-3}$	$1.18 \pm 0.24 \times 10^{-2}$
4	$\mathbf{8.02 \pm 1.86} \times 10^{-3}$	1.00 ± 0.00	$8.18 \pm 1.29 \times 10^{-1}$	$5.74 \pm 0.45 \times 10^{-1}$
6	$\mathbf{1.01 \pm 0.53} \times 10^{-1}$	1.00 ± 0.00	1.00 ± 0.02	$9.39 \pm 0.11 \times 10^{-1}$

All methods use four hidden layers of width $H = 128$. Trainable real degrees of freedom: Complex sinh=100,098; SIREN=50,049; mFF-PINN=66,305; MscaleDNN-2-sin=150,147.

Tables 4.1 and 4.2 report the final accuracies, and Figures 4.1 and 4.2 show the corresponding histories. In two dimensions the mean error of Complex sinh grows from 3.36×10^{-4} at order two to 1.01×10^{-1} at order six, and remains below all three baselines throughout; the margin is largest at order four, where the mean error is smaller than the next lowest mean by a factor of about 72. The same ordering holds in three dimensions at orders two and four. At order six in three dimensions, Complex sinh reaches 9.82×10^{-1} while SIREN reaches approximately 1 and the other two baselines are also near unit error, so this setting exhibits budget-limited saturation rather than a meaningful separation.

TABLE 4.2
Polyharmonic benchmark, $d = 3$: mean $\pm$ sample standard deviation of the held-out relative L^2 error over five seeds after 1200 s. The lowest mean in each row is set in bold.

$2m$	Complex sinh	SIREN	mFF-PINN	MscaleDNN-2-sin
2	$\mathbf{3.29 \pm 0.39} \times 10^{-3}$	1.00 ± 0.00	$4.61 \pm 1.39 \times 10^{-1}$	$2.33 \pm 0.25 \times 10^{-1}$
4	$\mathbf{1.57 \pm 0.73} \times 10^{-1}$	1.00 ± 0.00	$9.95 \pm 0.04 \times 10^{-1}$	$9.38 \pm 0.35 \times 10^{-1}$
6	$\mathbf{9.82 \pm 0.09} \times 10^{-1}$	1.00 ± 0.00	1.00 ± 0.01	1.07 ± 0.07

All methods use four hidden layers of width $H = 128$. Trainable real degrees of freedom: Complex sinh=100,354; SIREN=50,177; mFF-PINN=66,305; MscaleDNN-2-sin=150,531.

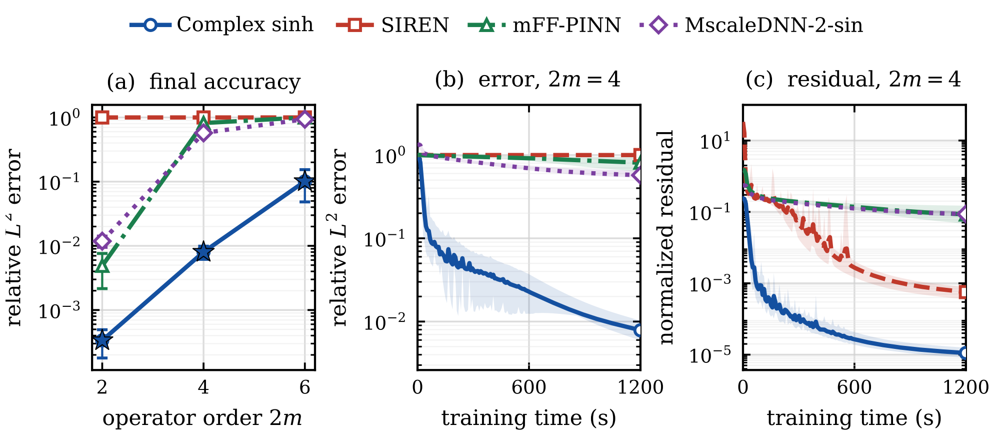

FIG. 4.1. *Two-dimensional polyharmonic benchmark. Panel (a) shows the five-seed mean relative L^2 error after 1200 seconds with sample-deviation bars, and a star marks the lowest mean at each order. Panels (b) and (c) show the seed-mean error and normalized interior residual histories of the representative fourth-order setting, with the seedwise range shaded.*

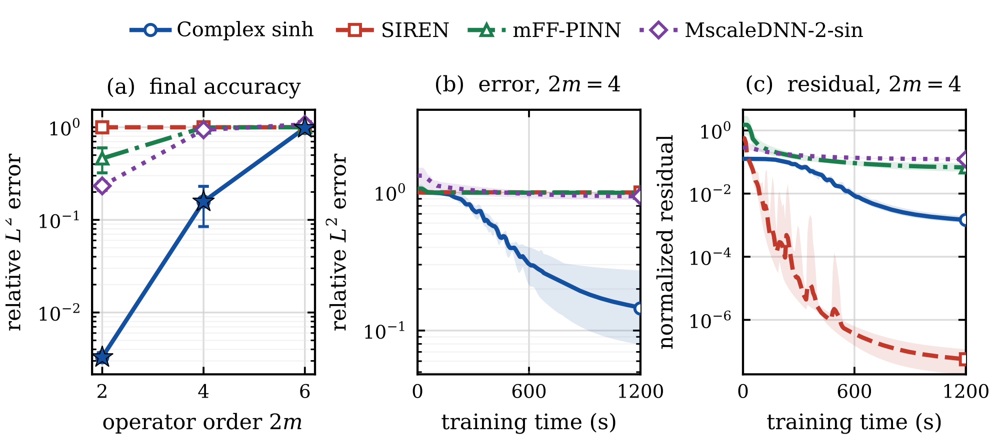

FIG. 4.2. *Three-dimensional polyharmonic benchmark, under the same reporting convention as Figure 4.1. The history panels use the fourth-order setting.*

4.4. Non-separable radial chirp. On $(-1,1)^2$ we solve the Helmholtz-type problem

$$-\Delta u + u = f, \qquad u_\star(x,y) = \sin\Big(\frac{a\pi}{2}(x^2+y^2)\Big), \tag{4.3}$$

with Dirichlet data taken from $u_\star$ and $a \in \{1,2,3\}$. Writing $\phi := a\pi(x^2+y^2)/2$ and $r^2 := x^2+y^2$, the manufactured source is

$$f = (a\pi)^2 r^2 \sin\phi - 2a\pi\cos\phi + \sin\phi. \tag{4.4}$$

Unlike a separable Fourier mode, this target has the local radial wavenumber $|\nabla\phi| = a\pi r$, so its frequency varies over the domain.

TABLE 4.3
Radial-chirp benchmark: mean ± sample standard deviation of the held-out relative L^2 error over five seeds after 1200 s. The lowest mean in each row is set in bold.

a	Complex sinh	SIREN	mFF-PINN	MscaleDNN-2-sin
1	$\mathbf{8.14 \pm 7.00} \times 10^{-4}$	$4.25 \pm 0.06 \times 10^{-1}$	$2.56 \pm 1.84 \times 10^{-3}$	$1.24 \pm 0.24 \times 10^{-2}$
2	$\mathbf{2.39 \pm 0.57} \times 10^{-3}$	1.53 ± 0.02	$9.69 \pm 1.00 \times 10^{-1}$	$8.47 \pm 1.85 \times 10^{-1}$
3	$\mathbf{1.00 \pm 0.18} \times 10^{-1}$	1.07 ± 0.02	1.42 ± 0.07	$9.02 \pm 0.97 \times 10^{-1}$

All methods use four hidden layers of width $H = 128$. Trainable real degrees of freedom: Complex sinh=100,098; SIREN=50,049; mFF-PINN=66,305; MscaleDNN-2-sin=150,147.

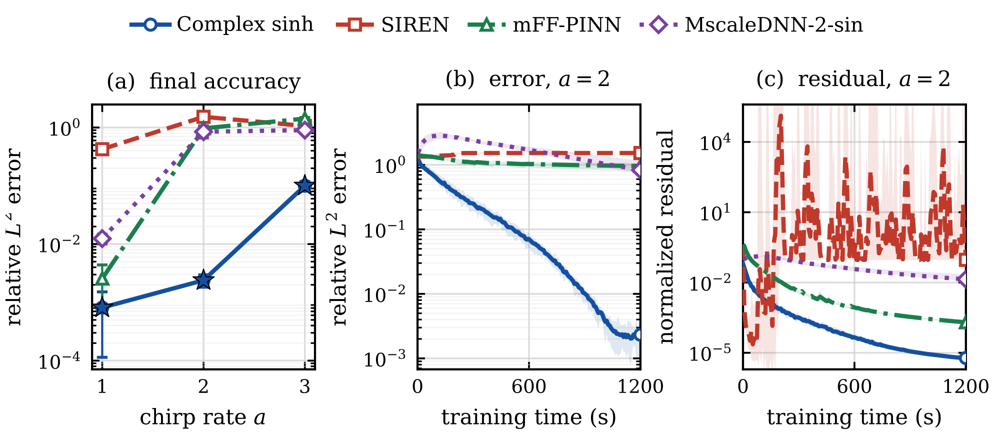

FIG. 4.3. *Radial-chirp benchmark, under the same reporting convention as Figure 4.1. The history panels use $a = 2$.*

Table 4.3 and Figure 4.3 collect the results, and Complex sinh attains the lowest mean error at each chirp rate. At $a = 1$ its mean error of 8.14×10^{-4} is smaller than that of mFF-PINN, the next most accurate method, by a factor of about 3. The separation grows at $a = 2$, where the three baselines have mean errors between 0.847 and 1.53 while Complex sinh remains at 2.39×10^{-3}. At $a = 3$ its error rises to 0.100 but stays below all three baselines. Since all four methods share one loss and one optimizer schedule, these numbers compare complete methods; by themselves they do not identify the value representation as the sole cause of the gap.

4.5. Time-harmonic Maxwell equations. For a two-dimensional transverse-magnetic mode, Maxwell's equations reduce to a scalar complex Helmholtz equation for the out-of-plane field E,

$$\Delta E + \kappa^2 E = f, \qquad \kappa^2 = (a\pi)^2(1 + \mathrm{i}\beta), \qquad E_\star = \exp\bigl(\mathrm{i}a\pi(x+y)\bigr), \tag{4.5}$$

with $\beta = 0.2$, exact Dirichlet data, $a \in \{2, 4, 6\}$, and the source $f = [-2(a\pi)^2 + \kappa^2]E_\star$. Complex sinh represents E natively, whereas each baseline uses a split real-imaginary pair of two width-128 networks.

TABLE 4.4

Time-harmonic Maxwell benchmark: mean ± sample standard deviation of the held-out relative L^2 error over five seeds after 1200 s. The lowest mean in each row is set in bold.

a	Complex sinh	SIREN	mFF-PINN	MscaleDNN-2-sin
2	$\mathbf{4.21 \pm 1.57} \times 10^{-3}$	1.00 ± 0.00	$6.93 \pm 1.86 \times 10^{-1}$	1.09 ± 0.06
4	$\mathbf{4.19 \pm 0.22} \times 10^{-2}$	1.00 ± 0.00	1.06 ± 0.11	1.29 ± 0.07
6	$\mathbf{1.20 \pm 0.17} \times 10^{-1}$	1.00 ± 0.00	1.13 ± 0.15	1.34 ± 0.04

All methods use four hidden layers of width $H = 128$. Trainable real degrees of freedom: Complex sinh=100,098; SIREN=100,098; mFF-PINN=132,610; MscaleDNN-2-sin=300,294.

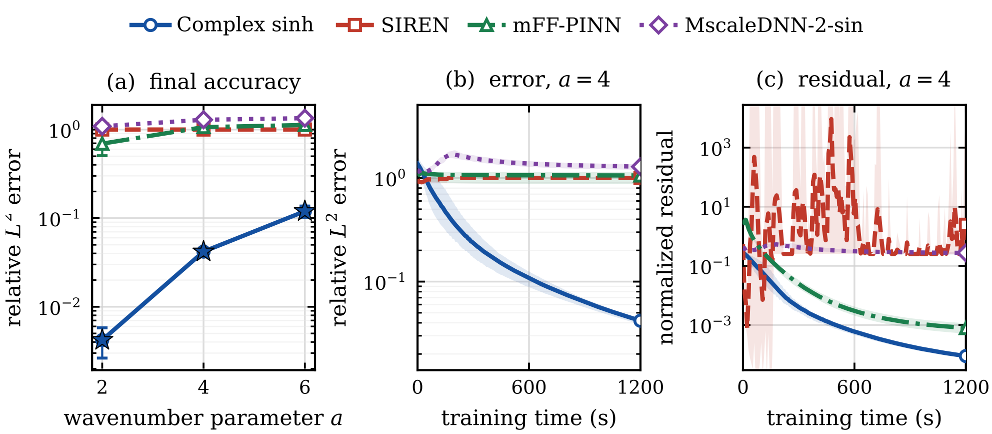

FIG. 4.4. *Time-harmonic Maxwell benchmark, under the same reporting convention as Figure 4.1. The history panels use $a = 4$.*

As Table 4.4 and Figure 4.4 show, the native-complex network attains the lowest mean error at every wavenumber, with mean errors growing from 4.21×10^{-3} at $a = 2$ to 0.120 at $a = 6$. The split-real baselines remain near or above unit error at the two higher wavenumbers. This benchmark favours a native complex representation by construction, and the comparison still contains architecture-dependent optimization and throughput effects, so it should not be read as a controlled ablation of complex arithmetic alone.

4.6. Throughput and aggregate outcome. The rows of Table 4.5 show why equal wall-clock time is a more informative control than an equal number of optimizer steps. On the scalar chirp task SIREN completes the fastest steps, while the branched

Table 4.5

Representative optimizer throughput: mean ± sample standard deviation of milliseconds per step over five seeds. Methods should be compared within a row; the stored software versions differ between settings.

setting	Complex sinh	SIREN	mFF-PINN	MscaleDNN-2-sin
Polyharmonic, $d = 2$, $2m = 4$	243.1 ± 0.2	243.7 ± 0.2	576.8 ± 0.2	729.7 ± 0.4
Radial chirp, $a = 2$	70.7 ± 0.2	61.9 ± 0.0	145.2 ± 0.1	182.3 ± 0.0
Maxwell, $a = 4$	70.8 ± 0.1	123.5 ± 0.0	290.1 ± 0.0	367.8 ± 0.1

mFF-PINN and the three MscaleDNN subnets are slower by factors of about 2 and 3. For Maxwell each split-real baseline evaluates two networks, which raises its per-step cost, and the polyharmonic row shows the cost of propagating fourth-order jets through branched architectures. The table stays within one setting per row because the recorded source commit differs between settings, so a throughput trend across a sweep would not be attributable to the sweep.

Across the 12 settings, Complex sinh attains the smallest five-seed mean error in every table. The margin to the next lowest mean exceeds a factor of 70 in four settings, namely the two-dimensional fourth-order and three-dimensional second-order polyharmonic problems, the radial chirp at $a = 2$, and Maxwell at $a = 2$; it is at least a factor of 3 in 11 of the 12 settings. The exception is the three-dimensional sixth-order polyharmonic setting, in which every method stays close to the error of the trivial predictor. The comparison does not isolate a single causal mechanism, since a complex model changes the value representation, the initialization, the parameter count, and the throughput at once: literal width $H = 128$ gives the scalar Complex sinh network exactly twice as many real degrees of freedom as SIREN, whereas MscaleDNN-2-sin has more degrees of freedom but completes fewer steps, and the loss weights and optimizer are shared rather than tuned per architecture. What the study supports is the claim it was designed for: an exact derivative backend built on the minimal schedule sustains a complete high-order and complex-valued workflow, at a per-step cost that leaves it competitive with methods tuned for such problems.

5. Conclusion. Evaluating one prescribed high-order mixed derivative exactly is a question about how few directions suffice, and we have shown that it is the Waring problem for the associated monomial. The identification is an equivalence rather than a bound: a directional formula of a given length exists precisely when the monomial decomposes into that many powers of linear forms. The rank formula of Carlini, Catalisano, and Geramita then evaluates the minimum over the complex numbers as the product of the incremented active exponents, and a roots-of-unity construction attains it in closed form, with coefficients known in advance rather than obtained from a linear solve. The classical polarization identity is the lowest-order case of that construction, which is why the gap between them opens only when an index repeats.

Combined with a batched Taylor jet through an entire activation and a closed-form vector-Jacobian product, the schedule becomes a derivative backend that is algebraically exact and differentiable in the model parameters. Double-precision tests reproduce nested automatic differentiation for values and for parameter gradients through order six. Used inside complete physics-informed training runs, a complex-parameter sinh network built on this backend attains the lowest five-seed mean relative L^2 error in all 12 settings of a 1200-second, width-128 protocol on polyharmonic, radial-chirp, and time-harmonic Maxwell problems, with a margin of at least a factor of 3 in 11 of them; the exception is the hardest three-dimensional sixth-order problem, on which every method saturates near the error of the trivial predictor.

The reduction is sharpest for repeated-index multi-indices and vanishes for square-free ones, where the rank coincides with the merged polarization count. Operator sums are handled term by term, so the construction carries no joint optimality claim for a sum, and trace-collapsing or stochastic methods remain preferable for a Laplacian or for a broad tensor contraction. Natural continuations are a rank-aware expansion of an operator sum that reuses directions across its terms, a conditioning analysis of the schedule at high order and in reduced precision, and the extension of the same apolarity argument to derivative tensors contracted against a fixed structured family.

Appendix A. Schedules, filters, and the apolar bound.

This appendix collects the roots-of-unity filter used in the proof of Theorem 3.3, the explicit schedules for the derivative patterns that occur in Section 4, and the apolarity argument behind the rank formula (3.10). The formulas of Appendix A.2 are those the implementation evaluates.

A.1. Roots-of-unity filter. LEMMA A.1. *For every integer $m \geq 1$ and every integer k,*

$$\sum_{\zeta \in U_m} \zeta^k = m\, \mathbf{1}_{m|k}. \tag{A.1}$$

Proof. Let $\omega := \exp(2\pi \mathrm{i}/m)$, so that every element of U_m is ω^ℓ for exactly one $\ell \in \{0, \dots, m-1\}$ and

$$\sum_{\zeta \in U_m} \zeta^k = \sum_{\ell=0}^{m-1} \left(\omega^k\right)^\ell.$$

If $m \mid k$, then $\omega^k = 1$ and the sum equals m. If $m \nmid k$, then $\omega^k \neq 1$, and the finite geometric series gives

$$\sum_{\ell=0}^{m-1} \left(\omega^k\right)^\ell = \frac{1-\omega^{km}}{1-\omega^k} = 0,$$

since $\omega^m = 1$. ∎

A.2. Explicit low-order schedules. The mixed terms of the polyharmonic residual (4.2) and the pure powers of the chirp and Maxwell residuals use the following instances of (3.11).

For a pure power $\partial^\nu = \partial_1^p$ we have $n = 0$, the schedule has one direction, and T_p is evaluated once,

$$\partial_1^p u(\boldsymbol{x}) = p!\, T_p(\boldsymbol{x}; e_1). \tag{A.2}$$

For the fourth-order mixed term $\partial^\nu = \partial_1^2 \partial_2^2$ we have $n = 1$, $a = (2, 2)$, and $m_1 = 3$; with $\omega := \exp(2\pi \mathrm{i}/3)$ and $U_3 = \{1, \omega, \omega^2\}$, formula (3.11) gives the three directions

$$\partial_1^2 \partial_2^2 u(\boldsymbol{x}) = \frac{4}{3} \sum_{\zeta \in U_3} \zeta\, T_p(\boldsymbol{x};\, e_1 + \zeta e_2), \tag{A.3}$$

against the four real directions retained by merged polarization.

For the sixth-order mixed term $\partial^\nu = \partial_1^2\partial_2^2\partial_3^2$ we have $n = 2$, $a = (2,2,2)$, and $m_1 = m_2 = 3$, so (3.11) gives $R_{\mathbb{C}}(\boldsymbol{z}^a) = 9$ directions,

$$\partial_1^2\partial_2^2\partial_3^2 u(\boldsymbol{x}) = \frac{8}{9}\sum_{\zeta_1,\zeta_2\in U_3} \zeta_1\zeta_2\, T_p\big(\boldsymbol{x};\, e_1 + \zeta_1 e_2 + \zeta_2 e_3\big), \tag{A.4}$$

against the 13 real directions of Table 3.1. The equivalent polynomial identity (3.9) for (A.4) reads

$$6!\, z_0^2 z_1^2 z_2^2 = \frac{8}{9}\sum_{\zeta_1,\zeta_2\in U_3} \zeta_1\zeta_2\,(z_0 + \zeta_1 z_1 + \zeta_2 z_2)^6, \tag{A.5}$$

and it can be checked directly: expanding the right-hand side and collecting the coefficient of $z_0^{\beta_0} z_1^{\beta_1} z_2^{\beta_2}$ produces the product $\prod_{j=1}^{2}\sum_{\zeta_j\in U_3}\zeta_j^{1+\beta_j}$, which by Lemma A.1 is nonzero only when $\beta_j \equiv 2 \pmod 3$ for $j = 1, 2$. Together with $\beta_0 + \beta_1 + \beta_2 = 6$ this forces $\beta = (2,2,2)$, and the surviving constant is $\frac{8}{9}\cdot\frac{6!}{2!\,2!\,2!}\cdot 9 = 720 = 6!$.

A.3. Apolar form of the rank bound. The lower bound in (3.10) comes from apolarity, which we recall for completeness. For a homogeneous $F \in \mathbb{C}[\boldsymbol{z}]_p$, let $F^\perp$ denote its apolar ideal, that is, the ideal of differential operators annihilating F, and call a finite subscheme $X \subset \mathbb{P}^n$ apolar to F when $I_X \subset F^\perp$. The Waring rank then admits the characterization

$$R_{\mathbb{C}}(F) = \min\big\{\deg X : X \subset \mathbb{P}^n \text{ finite, reduced, and apolar to } F\big\}. \tag{A.6}$$

For the monomial $F = \boldsymbol{z}^a$ the apolar ideal is the complete intersection

$$(\boldsymbol{z}^a)^\perp = \big(\partial_0^{a_0+1}, \ldots, \partial_n^{a_n+1}\big), \tag{A.7}$$

and the Hilbert function of the quotient $\mathbb{C}[\partial]/(\boldsymbol{z}^a)^\perp$ bounds the degree of an apolar subscheme from below by the product $\prod_{j=1}^n m_j$ of (3.10); see [11, 13] for the general theory and [2] for the monomial case. The construction (3.11) exhibits a reduced apolar subscheme of exactly that degree, so the bound is attained.

REFERENCES

[1] Jesse Bettencourt, Matthew J. Johnson, and David Duvenaud, *Taylor-mode automatic differentiation for higher-order derivatives in JAX*, in NeurIPS Workshop on Program Transformations for Machine Learning, 2019.

[2] Enrico Carlini, Maria Virginia Catalisano, and Anthony V. Geramita, *The solution to the Waring problem for monomials and the sum of coprime monomials*, Journal of Algebra, 370 (2012), pp. 5–14.

[3] Pao-Hsiung Chiu, Jian Cheng Wong, Chinchun Ooi, My Ha Dao, and Yew-Soon Ong, *CAN-PINN: A fast physics-informed neural network based on coupled-automatic–numerical differentiation method*, Computer Methods in Applied Mechanics and Engineering, 395 (2022), p. 114909.

[4] Nicholas Gao, Jonas Köhler, and Adam Foster, *folx: Forward Laplacian for JAX*, 2023. Software.

[5] Andreas Griewank and Andrea Walther, *Evaluating Derivatives: Principles and Techniques of Algorithmic Differentiation*, Society for Industrial and Applied Mathematics, Philadelphia, PA, 2nd ed., 2008.

[6] Jiequn Han, Arnulf Jentzen, and Weinan E, *Solving high-dimensional partial differential equations using deep learning*, Proceedings of the National Academy of Sciences, 115 (2018), pp. 8505–8510.

[7] Di He, Shanda Li, Wenlei Shi, Xiaotian Gao, Jia Zhang, Jiang Bian, Liwei Wang, and Tie-Yan Liu, *Learning physics-informed neural networks without stacked back-propagation*, in Proceedings of the 26th International Conference on Artificial Intelligence and Statistics, vol. 206 of Proceedings of Machine Learning Research, PMLR, 2023, pp. 3034–3047.

[8] Zheyuan Hu, Zekun Shi, George Em Karniadakis, and Kenji Kawaguchi, *Hutchinson trace estimation for high-dimensional and high-order physics-informed neural networks*, Computer Methods in Applied Mechanics and Engineering, 424 (2024), p. 116883.

[9] ———, *Bias-variance trade-off in physics-informed neural networks with randomized smoothing for high-dimensional PDEs*, Journal of Computational Physics, 521 (2025), p. 113524.

[10] Zheyuan Hu, Khemraj Shukla, George Em Karniadakis, and Kenji Kawaguchi, *Tackling the curse of dimensionality with physics-informed neural networks*, Neural Networks, 176 (2024), p. 106369.

[11] Anthony Iarrobino and Vassil Kanev, *Power Sums, Gorenstein Algebras, and Determinantal Loci*, vol. 1721 of Lecture Notes in Mathematics, Springer, Berlin, 1999.

[12] George Em Karniadakis, Ioannis G. Kevrekidis, Lu Lu, Paris Perdikaris, Sifan Wang, and Liu Yang, *Physics-informed machine learning*, Nature Reviews Physics, 3 (2021), pp. 422–440.

[13] Joseph M. Landsberg, *Tensors: Geometry and Applications*, vol. 128 of Graduate Studies in Mathematics, American Mathematical Society, Providence, RI, 2012.

[14] Ruichen Li, Haotian Ye, Du Jiang, Xuelan Wen, Chuwei Wang, Zhe Li, Xiang Li, Di He, Ji Chen, Weiluo Ren, and Liwei Wang, *A computational framework for neural network-based variational Monte Carlo with Forward Laplacian*, Nature Machine Intelligence, 6 (2024), pp. 209–219.

[15] Ziqi Liu, Wei Cai, and Zhi-Qin John Xu, *Multi-scale deep neural network (MscaleDNN) for solving Poisson–Boltzmann equation in complex domains*, Communications in Computational Physics, 28 (2020), pp. 1970–2001.

[16] Maziar Raissi, Paris Perdikaris, and George E. Karniadakis, *Physics-informed neural networks: A deep learning framework for solving forward and inverse problems involving nonlinear partial differential equations*, Journal of Computational Physics, 378 (2019), pp. 686–707.

[17] Zekun Shi, Zheyuan Hu, Min Lin, and Kenji Kawaguchi, *Stochastic Taylor derivative estimator: Efficient amortization for arbitrary differential operators*, Advances in Neural Information Processing Systems (NeurIPS), 37 (2024), pp. 122316–122353.

[18] Vincent Sitzmann, Julien N. P. Martel, Alexander W. Bergman, David B. Lindell, and Gordon Wetzstein, *Implicit neural representations with periodic activation functions*, Advances in Neural Information Processing Systems (NeurIPS), 33 (2020), pp. 7462–7473.

[19] Charles M. Stein, *Estimation of the mean of a multivariate normal distribution*, The Annals of Statistics, 9 (1981), pp. 1135–1151.

[20] Matthew Tancik, Pratul P. Srinivasan, Ben Mildenhall, Sara Fridovich-Keil, Nithin Raghavan, Utkarsh Singhal, Ravi Ramamoorthi, Jonathan T. Barron, and Ren Ng, *Fourier features let networks learn high frequency functions in low dimensional domains*, Advances in Neural Information Processing Systems (NeurIPS), 33 (2020), pp. 7537–7547.

[21] Sifan Wang, Yujun Teng, and Paris Perdikaris, *Understanding and mitigating gradient flow pathologies in physics-informed neural networks*, SIAM Journal on Scientific Computing, 43 (2021), pp. A3055–A3081.